Nuthanan Tharmarajah

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Education

University of Waterloo — Waterloo, ON
Bachelor of Software Engineering — *Aug. 2024 – May 2029*

- **GPA:** 3.9/4.0
- **Activities and Societies**: UW Formula Electric, SE Yearbook Representative, Tamil Student Association

Technical Skills

Languages: Swift, Java, Golang, Python, C/C++, JavaScript, TypeScript, HTML/CSS
Frameworks: React, Node.js, TailwindCSS, Bootstrap, Flask, Next.js, FastAPI, Express, React Native, SwiftUI
Cloud/Tools: GCP (APIs, OCR, Storage), AWS (EC2, Lambda, CloudFront), Docker, Git, CI/CD, 1Password
Databases: Postgres, SQLAlchemy, MongoDB, Firebase

Experience

Software Development Engineer — May 2025 – Present
Guelph Eye MD — *Guelph, ON*

- Collaborated with medical staff to automate **10+ staff workflows** with **Golang**, cutting task time by **30%**
- Built a **fax automation tool** with a **Golang backend** and **Electron desktop app** to display parsed fields, integrating **StreamDecks** for workflow control and **GCP OCR** for fax parsing, reducing manual work by **90%**
- Designed **Postgres databases** for **3,000+ clinic events**, allowing for long-term analytics and data analysis
- Refactored backend & **CI/CD**, added **1Password** secrets, ensured **HIPAA** compliance, boosting reliability
- Developed a **Raspberry Pi** scanner in **Python** to automate patient check-ins, reducing reception workload

Lead Website Developer — May 2025 – Present
SSICSIM 2025 — *Toronto, ON*

- Built the official SSICSIM website using **React and TailwindCSS**, improving usability for **300+ delegates**
- Established a **feedback loop with 80+ staff**, applying user-centered design principles to iteratively refine UI
- Optimized website performance and reliability by integrating the **AWS CloudFront CDN** for images and fonts, designing for scalable load distribution and fast page delivery across **8,000+ monthly pageviews**

Lead Website Developer — June 2024 — August 2024
K&T Construction Service — *Toronto, ON*

- Developed a responsive website using **HTML and CSS** to showcase the company's services and projects
- Engaged with **20+ staff** to gather website application requirements **through surveys and interviews**, analyzed feedback to guide design decisions, ensuring alignment with client needs, improving website engagement by **47%**
- Increased website traffic by **30%** over three months by optimizing load times and SEO using **Google Analytics**

Projects

Solana Swift SDK | *Swift, Solana JSON-RPC, Web3* — December 2025
Grant-funded by Solana Foundation

- Built a **native Swift SDK** (**3,800+ lines**) enabling iOS/macOS apps to integrate **3+ Solana wallets** (Phantom, Solflare, Backpack), supporting session establishment, message signing, and transaction execution
- Designed a **type-safe transaction and instruction builder** using async/await Solana JSON-RPC with support for **System, Token, Associated Token, and Memo programs**, mirroring established Solana SDK interfaces
- Authored **documentation** and wrote **400+ lines of tests** to validate wallet connections and signing flows

Solshare | *Swift, Firebase, Cohere API, Solana, Typescript* — September 2025
Winner at Hack the North 2025 — Best Use of Cohere API & Best Use of Solana

- Built an **iOS UI in Swift** to simplify receipt tracking, item assignment, and dispute resolution for friends
- Integrated **Firebase with WebSocket-based real-time sync** for instant updates to group expenses, balances, and shared activity, and connected **Solana** to enable secure, zero-trust payments between users
- Built an **LLM pipeline with Cohere APIs** using Command + Vision and reasoning models to itemize receipts